

Database Systems

Project Report

Members:

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1) Title

Airline Reservation System

2) Introduction (Aim or Motivation)

The aim of this project is to design and develop an Airline Reservation System that allows passengers to easily book, view, and cancel flight tickets online. The system also provides an admin panel to manage flights, bookings, and financial records. The motivation behind this project is to replace manual booking systems with a fast, reliable, and user-friendly digital system. It helps passengers save time and allows airlines to manage operations efficiently.

3) Background (Research and Project Selection)

Airline reservation systems are widely used in the travel industry to manage flight operations and bookings efficiently. Popular systems like Amadeus GDS and Sabre are used worldwide to handle flight reservations and related services. During the research phase, it was observed that such systems typically include key functionalities such as flight scheduling, ticket booking, payment processing, and customer management. This project was selected because it covers important database concepts, including data storage, relationships between entities, transaction handling, and user authentication. These concepts are essential for understanding how real-world database systems operate and manage large amounts of data efficiently. Additionally,

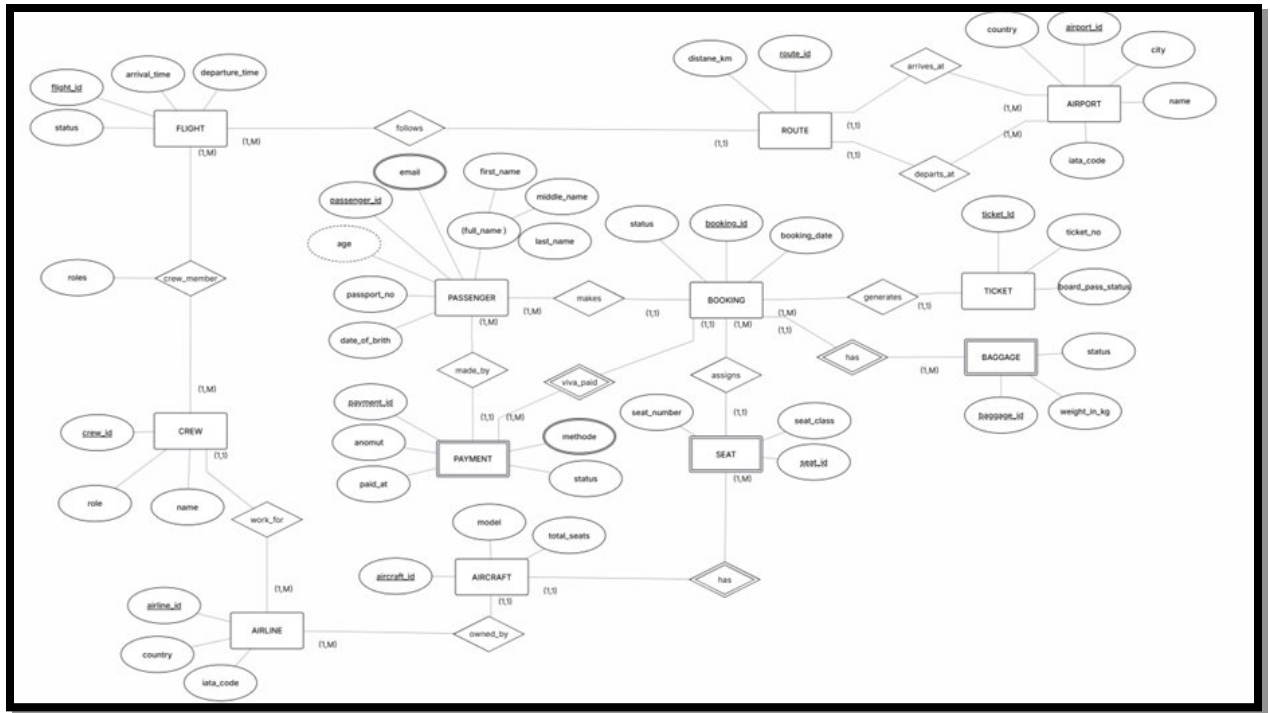
this project helps in developing practical skills in designing and implementing a complete database-driven application.

4) Project Specification

The system developed in this project is named Airline Reservation System. It is designed to serve two main types of users: Passenger and Admin. The passenger can perform activities such as searching for flights, booking tickets, selecting seats, making online payments, viewing tickets, and canceling bookings. On the other hand, the admin is responsible for managing the overall system, including viewing all bookings, updating flight details, canceling or rescheduling flights, and monitoring the financial performance of the airline. The system is developed using web technologies. The frontend is created using HTML and CSS to provide a simple and user-friendly interface. The backend is developed using Python with the Flask framework, which handles the application logic and communication between the user interface and the database. The database is managed using MySQL, where all data such as passenger details, flight information, bookings, and payments are stored and managed efficiently. The system consists of several main modules, including User Authentication, which provides secure login for both passengers and admins; Flight Search and Booking, which allows users to find and reserve flights; Payment System, which manages online transactions; and the Admin Dashboard, which enables full control over bookings, flights, and system operations.

5) Information of ERD

The Airline Reservation System consists of several main entities, including Passenger, Admin, Flight, Booking, Payment, and Ticket. Each of these entities plays an important role in managing the system's operations. The Passenger entity stores user information and is connected to the Booking entity through a one-to-many relationship, meaning one passenger can make multiple bookings. Each Booking is associated with exactly one Flight, which represents a many-to-one relationship, as multiple bookings can be made for the same flight. Furthermore, each booking has a one-to-one relationship with the Payment entity, ensuring that every booking corresponds to a single payment transaction. The Flight entity is also linked to the Ticket entity in a one-to-many relationship, where one flight can have multiple tickets issued for different passengers. These relationships help maintain proper data organization and ensure efficient management of airline operations within the system.



6) Normalization

$Flight_Crew (Flight_id, Crew_id, role \{ Nome \})$

1NF

already in 1NF
No multi-value
No repeating groups

2NF

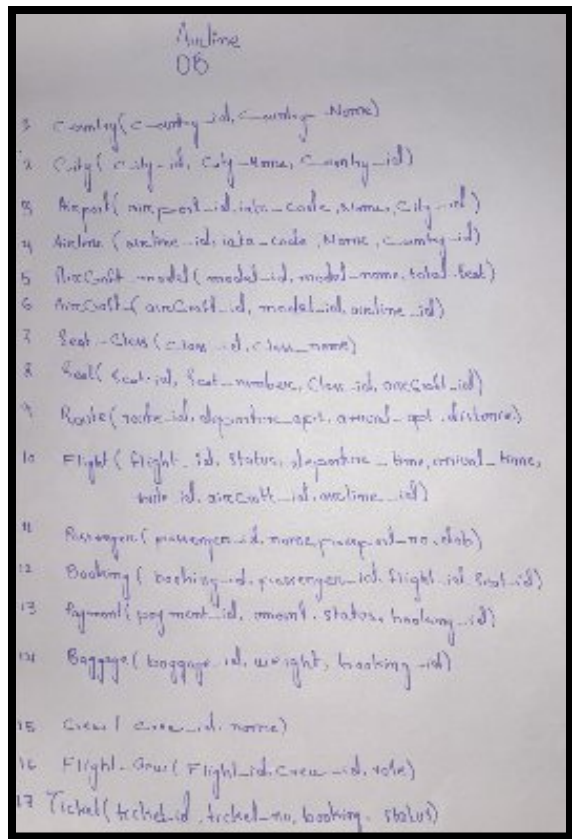
Composite key $(Flight_id, Crew_id)$
Partial dependence
 $Crew_id \rightarrow Nome$
 $(Flight_id, Crew_id) \rightarrow role$

3NF

No transitive dependency

Hence

$Flight_Crew (Flight_id, Crew_id, role)$
 $Crew (Crew_id, Nome)$



7) Problem Analysis

Traditional airline booking systems have several limitations that affect both passengers and administrators. In many cases, manual booking processes are slow and time-consuming, which reduces overall efficiency. Data management becomes difficult due to large volumes of records, and there is a high chance of errors during data entry or processing. Additionally, there is often no proper system for tracking the financial performance of the airline, making it difficult for management to analyze revenue and expenses accurately. Users also face multiple issues such as difficulty in booking tickets, lack of proper seat selection options, and the absence of an easy and quick cancellation system. These problems highlight the need for a more efficient, automated, and user-friendly airline reservation system.

8) Solution Design (Project Details, Functionality and Features)

The proposed system solves the limitations of traditional airline booking systems by providing a complete digital solution that improves efficiency and user experience.

For passengers, the system allows users to register and log in securely, search for available flights, book tickets, select seats, make online payments, view their tickets, and cancel bookings easily. For the admin, the system provides powerful management features such as viewing all bookings, managing flights by adding, updating, or deleting them, canceling or rescheduling flights, and monitoring the financial performance of the airline. In terms of security, the system ensures separate authentication for passengers and admins, along with a secure login mechanism to protect user data and maintain system integrity.

9) Implementation and Testing Phase

The implementation phase of the Airline Reservation System involved creating a structured and efficient database using MySQL. All tables were designed and implemented according to the ERD to ensure proper relationships between entities such as passengers, flights, bookings, and payments. The backend logic was developed using Python with the Flask framework to handle user requests, process data, and connect the frontend with the database. The frontend interface was designed using HTML and CSS to provide a simple and user-friendly experience for both passengers and admin users. The testing phase was carried out to ensure that each module of the system worked correctly. Unit testing was performed on all individual components, including authentication, booking, and payment modules. The login system was tested to verify secure access for both passenger and admin users. Booking and cancellation features were also tested to ensure proper functionality, and the payment system was verified for successful transactions. Any errors found during testing were fixed to improve system performance and ensure smooth and reliable operation.

10) Project Breakdown Structure

The implementation phase of the project involved several important steps to ensure the proper functioning of the Airline Reservation System. The database was created using MySQL, and all tables were designed according to the ERD to maintain proper relationships between entities such as passengers, flights, bookings, and payments. The backend was developed using Python with the Flask framework, which handled the application logic and connected the frontend with the database. The frontend interface was designed using HTML and CSS to provide a simple and user-friendly experience for users. The testing phase was carried out to ensure that all modules of the system worked correctly and efficiently. Unit testing was performed for each

module, including authentication, booking, and payment systems. The login system was tested to ensure secure access for both passengers and admins. Additionally, booking and cancellation features were tested to verify their correct functionality, while the payment system was checked to ensure successful transaction processing. Any errors identified during testing were fixed to improve system performance and ensure smooth operation.

11) Results

User Account Creation

The screenshot shows the 'Create your account' page on the SkyWay website. The page has a dark blue header with the SkyWay logo and 'Sign in' and 'Create account' links. The main content area is a light beige color. The form is titled 'Create your account' with the subtitle 'Just a few details and you're ready to fly.' It is divided into two sections: 'ACCOUNT' and 'TRAVELER DETAILS'. The 'ACCOUNT' section has fields for 'USERNAME *' (filled with 'ALI AHMED'), 'PASSWORD *' (filled with '*****'), and 'CONFIRM PASSWORD *' (filled with '*****'). The 'TRAVELER DETAILS' section has fields for 'FULL NAME *' (filled with 'ALI AHMED'), 'PASSPORT NO. *' (filled with '123456789'), and 'DATE OF BIRTH *' (filled with '03-May-2026'). There is a calendar icon next to the date field. At the bottom of the form is an orange 'Create account' button and a link 'Already a member? Sign in'.

User Dashboard

The screenshot shows the 'User Dashboard' on the SkyWay website. The page has a dark blue header with the SkyWay logo and navigation links: 'Home', 'Search Flights', 'My Bookings', 'My Profile', 'Check-in', 'Notifications', 'ALI AHMED', and 'Sign out'. The main content area is a light beige color. At the top, there is a green banner that says 'Welcome aboard, ALI AHMED!'. Below the banner is a dark blue box with the title 'Where would you like to fly?' and the subtitle 'Search across 7 airports and book your seat in seconds.' It contains three input fields: 'FROM' (filled with 'Any airport'), 'TO' (filled with 'Any airport'), and 'DEPARTURE DATE' (filled with 'dd-YYYY'). There is a calendar icon next to the date field and an orange 'Search flights' button. Below the search box are four white cards: 'My Bookings' (with a calendar icon and text 'View, modify, or cancel'), 'Online Check-in' (with a checkmark icon and text 'Get your boarding pass'), 'My Profile' (with a person icon and text 'Personal details'), and 'Notifications' (with a bell icon and text 'Updates & alerts'). At the bottom, there are four white cards showing statistics: 'SCHEDULED FLIGHTS' (5), 'MY BOOKINGS' (0), 'MY TOTAL BOOKINGS' (0), and 'SPENT (PKR)' (0). Each card has a small icon and a green checkmark.

User Search Flights Interface

The screenshot displays the 'SkyWay' flight search interface. At the top, a navigation bar includes links for Home, Search Flights, My Bookings, My Profile, Check-In, Notifications, and a user profile for ALI AHMED with a Sign out option. The main section is titled 'Search flights' and contains a search form with fields for 'FROM' (Any airport), 'TO' (Any airport), and 'DATE' (dd-mm-yyyy). Below the form are 'Search flights' and 'Reset' buttons. A section titled 'Available flights (7)' contains a table with the following data:

FLIGHT	AIRLINE	AIRCRAFT	FROM	TO	DEPARTURE	ARRIVAL	SEATS	STATUS
F1	PIA	Boeing737	KHI (Karachi)	DXB (Dubai)	2025-06-01 08:00	2025-06-01 10:00	179 / 180	Scheduled view book
F2	Emirates	A380	KHI (Karachi)	LHR (London)	2025-06-02 09:00	2025-06-02 16:00	499 / 500	Scheduled view book
F3	Emirates	Boeing777	DXB (Dubai)	JFK (New York)	2025-06-03 10:00	2025-06-03 18:00	349 / 350	Delayed view book
F4	British Airways	A320	LHR (London)	JFK (New York)	2025-06-04 11:00	2025-06-04 14:00	149 / 150	Scheduled view book
F5	Qatar Airways	A350	JFK (New York)	DOH (Doha)	2025-06-05 08:00	2025-06-05 09:00	299 / 300	Cancelled view

12) Conclusion

The Airline Reservation System was successfully developed and tested, providing an efficient and reliable way to manage flight bookings and airline operations. The system significantly improves the speed of booking, ensures better data accuracy, and enhances the overall user experience for both passengers and administrators. In the future, the system can be further improved by adding mobile application support, integrating real-time flight tracking, and strengthening security features to make it more robust and user-friendly. Overall, this project demonstrates a strong understanding of database systems, web application development, and the practical implementation of an airline reservation system.